What the project is doing

From approved paired stimuli to a controlled test of visual context and motor-related representations.

The question

Does body continuity change how a model represents observed finger movement? The intended comparison is a static sequence (A to A) against movement (A to B), shown in three body contexts.

What this repository delivers

A portable copy of the final dataset-expansion method, frozen local inputs, exact pixel checks and a human review gate. It expands already approved hand pairs; it does not synthesize new movement.

20
approved source pairs

160
diffusion desk parents

480
appearance families

2,880
condition PNGs

One family = one source + desk + seed + white balance. It contains static/movement frames in full body, upper limb and detached hand contexts. All saved PNGs remain 2048 x 2048; hidden context pixels are black.

Current status: the original 480 families were accepted on 11 October 2026, with a reference-17 clothing flag. Their images remain unchanged. Desk masks were corrected after acceptance (page 4); these new inputs need a new canary. The dataset has 20 source groups with 24 appearances each. Scientific replication has not been validated.

The earlier TRDP-I report explored CLIP/DINO representations and proposed motor-response modeling. Its older 252-family dataset and findings are separate from this 480-family production snapshot.

Current production recipe

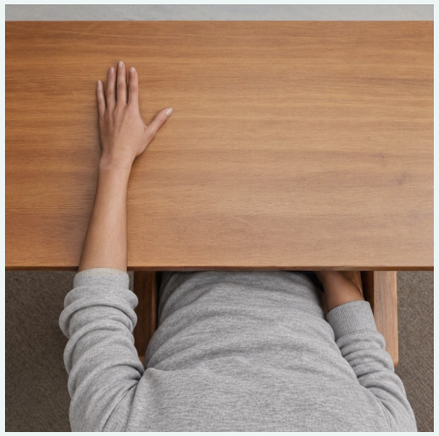
Everything below describes the final desk-expansion stage. Earlier hand and clothing repairs are already frozen in the source images.

1. Reuse approved V7 inputs	20 V7 A/B pairs. Reuse frozen 1024-square initial images and Canny controls (30/75). Corrected native desk masks place each cropped hand matte back in its scene crop, with a 36 px guard and 48 px feather (page 4). Reuse these stored inputs; do not rerun legacy preparation.
2. Make one desk proposal	RealVisXL V4.0 + Xinsir Canny SDXL + fixed fp16 SDXL VAE. Render 1024 x 1024 in fp16; DPM++ multistep / Karras. 40 requested steps, strength 0.4 = 16 effective steps; CFG 3.5; Canny scale 0.65, control end 0.9. Same prompt in both encoders, each at most 77 tokens.
3. Share protected composition	Enlarge with Lanczos to 2048. Match the desk-core median color, clipping the RGB offset to +/-24. Blend only by the frozen desk weight. Copy exactly the same edited desk into A and B. The editable mask must not overlap the original action difference.
4. Expand appearance	Beech, oak, maple and birch; seeds 9 and 123. Shared linear-RGB gains: neutral [1, 1, 1], warm [1.035, 1, 0.965], cool [0.965, 1, 1.035]. No new shadow geometry or spatial transform.
5. Export and check	Full-body, upper-limb and detached-hand views, each with static and movement frames. Reopen saved PNGs: check hashes, RGB mode, 2048-square size, exact retained pixels and black hidden regions. Pair differences must remain inside the frozen action support.
6. Review before scaling	Ten canary families need recorded visual approval before a full run. Check texture, seams, contact, anatomy/clothing, action and lighting. A numeric pass never certifies realism. Historical approval does not approve new generated outputs.

Worked example / input to shared desk

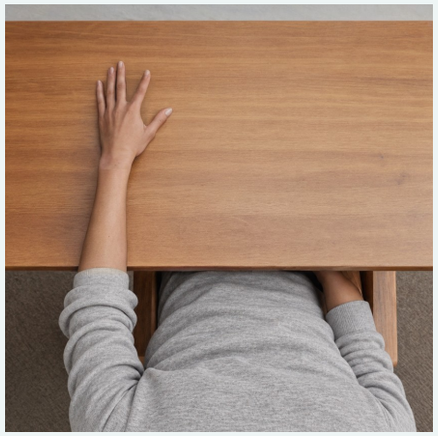
Reference 6, birch, seed 9. Panels 4-5 are historical outputs made with the old mask; panel 2 shows the corrected input for a future canary.

1A Frozen static A



Input: approved scene and hand.

1B Frozen movement B



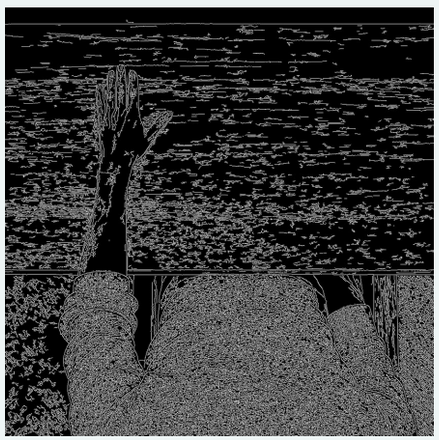
Input: approved index action.

2 Corrected desk weight



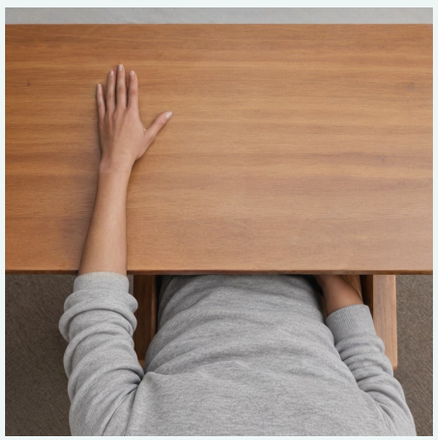
Future input; white edits, black protects.

3 Frozen Canny control



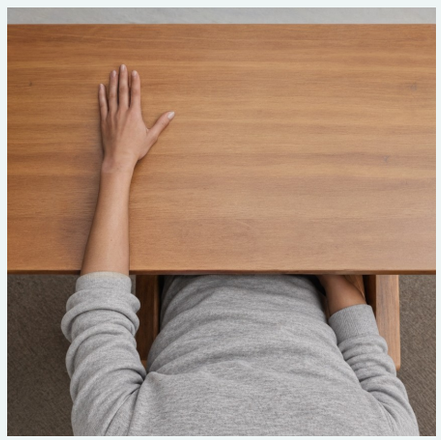
1024-square guide; threshold 30/75.

4 Historical SDXL proposal



Old-mask run; 16 steps at 1024.

5 Historical shared composite



Old-mask result; no regeneration here.

Only weighted desk pixels enter the native composition, shared between A and B. The historical result above has not been recomposited with the corrected mask. Its approval does not approve a future corrected run.

Mask correction / put the crop back

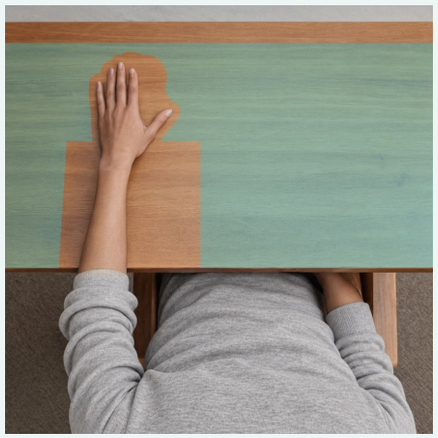
All 20 masks had the same coordinate error. The oversized hand silhouette came from stretching a cropped hand matte over the entire scene.

BEFORE / misplaced protection



Cyan edits desk; the large brown hand is wrong.

AFTER / correctly placed protection



Cyan now reaches the desk beside the hand.

BEFORE / historical desk weight



Old weight; retained only as evidence here.

AFTER / corrected desk weight



Current local input; gray softens the boundary.

1. Restore native coordinates

The hand matte is a 1024 x 1024 crop, not a full-scene image. For reference 6, resize it to 583 x 583 and paste at (323, 207) on the 2048 x 2048 scene. The crop ends at (906, 790).

2. Read the protected area

Black protects the hand in both poses, a 36 px safety guard, the forearm/contact corridor and all non-desk pixels. It is deliberately wider than the visible hand. White allows desk editing. Cyan marks editable pixels in the photo overlays.

3. Understand the gray edge

The binary mask has only black and white. The blending weight adds a 48 px ramp inside the editable area: black keeps the source, white uses the new desk, gray mixes them. Neither file is shown as a stimulus.

4. Keep the historical batch

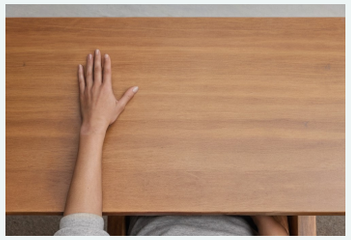
The accepted 480 families and their hashes are unchanged. They used the old masks. Corrected inputs require a fresh output folder and visual canary; old acceptance cannot approve them.

Correction checks: all 20 reconstructed old weights matched exactly; corrected weights never overlap the native hand matte or A/B action difference. Historic outputs must be checked with historic inputs or their accepted hashes.

Worked example / historical six outputs

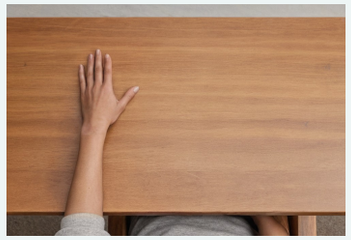
Original reference-6 results, made with the old masks. The three lighting transforms are shared; the six images below use the warm branch.

6 Neutral



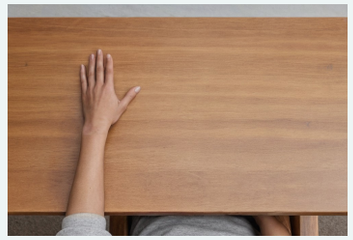
[1.0, 1.0, 1.0]

6 Warm



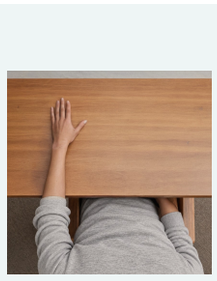
[1.035, 1.0, 0.965]

6 Cool



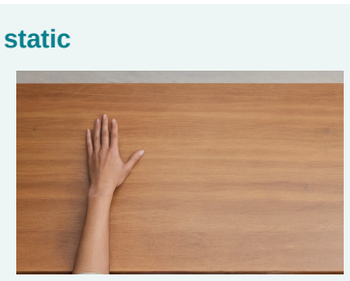
[0.965, 1.0, 1.035]

7 full body / static



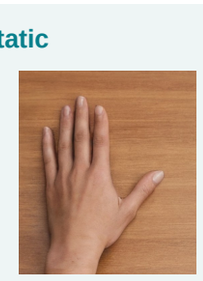
A to A

7 upper limb / static



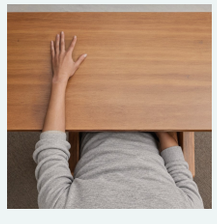
A to A

7 detached hand / static



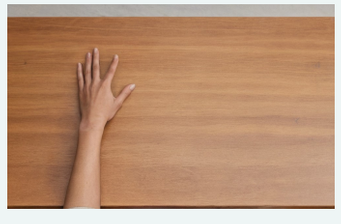
A to A

7 full body / movement



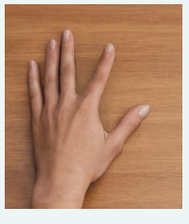
A to B

7 upper limb / movement



A to B

7 detached hand / movement



A to B

These historical exports remain unchanged. Previews remove black padding; saved PNGs keep the 2048-square canvas. Static trials show A twice; movement trials show A then B. A corrected-mask canary has not been generated.

What was tried / model and pipeline choices

Grouped trial ledger: variants are named together to keep the report short. A superseded screen is not evidence that a method can never work.

Unguided scene generation	DreamShaper SD1.5 (768) and DreamShaperXL v2 Turbo (1024) were compared. Unguided anatomy and composition were unreliable; SDXL became the base for guided experiments.
Control types	Canny, MiDaS depth, OpenPose, SegFormer-B5 segmentation and PidiNet soft edges were tested. First-person pose and segmentation were unreliable; soft edges gave weaker boundaries. Canny was retained. Canny + depth weakened edges and was about 33% slower in the earlier comparison.
Checkpoint comparison	SDXL Base, JuggernautXL v9, NightVisionXL, RealVisXL V4, Pony Realism v2.3, epiCRealism SD1.5, Realistic Vision 5.1/6 and SD3.5 Medium were screened. RealVis best fit the selected photorealism tradeoff. FLUX.1 dev hit the 16 GB memory limit; this is not an image-quality rejection.
Parameter and scheduler screens	Earlier grids varied CFG 1-9, Canny scale 0.3-1.1, control end 0.6-1.0 and 10-60 steps; Euler, DPM++ Karras, UniPC and DDIM were compared. The older CFG 4 / scale 0.9 / end 0.8 / 30-step recipe was superseded by the final desk recipe on page 2.
Appearance and prompt screens	Original/scene prompts, blurred/high-threshold Canny, img2img, SDXL Base and early control stop: high thresholds lost boundaries; early stop bent contours. DPM++ was useful. Later baseline/detailed/detailed-negative/dual-encoder screens scored 14/16, 16/16, 16/16 and 14/16 numeric passes; these small screens do not establish general visual yield.
Enlargement and refinement	Real-ESRGAN x2/x4, BSRGAN, Swin2SR x4, SD x4 Upscaler, ControlNet Tile x2 and SDXL hires fix were compared. Hires fix was used earlier; final expansion uses Lanczos to protect reviewed anatomy. Uncontrolled, controlled and 80-nominal-step refinement gave little benefit over Lanczos; tile/Swin2SR variants could degrade structure.
Original movement edits	Rectangular composites, wider masks, whole-hand crops, disabled registration, dedicated inpaint (baseline/stronger/lower-control/combined), no ControlNet, erase-redraw, lower control and DPM++ were tried, including holdouts. Failures included protected-pixel changes, ghosts, neighboring-digit changes and translucency. Collar tone and Poisson blending did not fix bad raw anatomy.

What was tried / learned editors and cutouts

Existing experiment evidence is shown below. These methods are not invoked by the final expansion code.

FoundHand reposing / masked sampling	12 completed trials across three recipes. Reposing altered bend, nails and shading even with the same target pose. Masked CFG 2.5 was best but under-opened the index (12.24 vs 17.25 degrees; 5.89 vs 9.34 in two cases). CFG 4.5 left remnants and produced little opening.
Full-noise SDXL and empty plates	Four dedicated nine-channel inpaint trials produced gray foreign material despite control. Whole-hand and narrow-index empty-desk trials invented boxes or tools. Neither became the movement method.
Qwen Image Edit 2511	Transformer-only NF4: two attempts ran out of memory before denoising. Transformer + text-encoder NF4: two 40-step runs completed with noisy gray images / visible pose colors. Full BF16 was not tested; do not call it a failed experiment.
ObjectClear and segmentation	20 ObjectClear calls: original masks left remnants, an alternate seed helped one case, and low CFG left peripheral remnants. Full-contour CFG 2.5 / seed 10031 was useful upstream. GrabCut misclassified shadows/gaps; converted SAM-HQ gave stripe masks despite high predicted IoU. Author SAM-HQ plus reviewed gap corrections was retained upstream, then frozen.
Layered desk and shadow variants	Texture donors (offset, blended, revised support) left seams. Hard cutouts left old slivers/web notches. Continuous-root shadow fits made forearm patches; inpainted stationary shadow retained ghosts; local shadow replacement made translucent brown web patches.



Archived rectangular edit: static parent (left), movement composite (right). Protected pixels changed in four of six screening outputs; this was rejected.



Archived Qwen NF4 run: input (left), actual completed output (right). Completion did not produce a usable stimulus.

What was tried / geometry and source repair

Useful upstream components are distinguished from rejected variants. None of these exploratory runners is needed to expand the frozen V7 sources.

Raster / rigid-layer variants	Opaque whole-contour rasterization made a skin triangle; half-opacity topology hit fold guards; whole-mesh vector roots made jagged extra skin. Rigid upper-finger + local-root variants retained a palm flap; removing that tail made a one-pixel crack. Opaque index + harmonic piecewise-affine root was a useful five-case upstream prototype, not a bulk acceptance guarantee.
Warp families and fresh pilots	Polar and continuous-root variants, sparse v1-v5, biharmonic v6/v7, harmonic v8, fair v9 and separator diagnostics were tested. Some reached 20/20 numerical checks, yet ownership, flaps, seams and realism still needed review. Fresh pilot v2 scored 14/20; v3/v4 scored 20/20 numeric passes. Numerical geometry did not certify the images.
V5 repairs / 120 recorded calls	Shared hand/nails and local movement repairs replaced the first movement cohort, rejected for rough knuckle texture. Revised masks and a seed-123 texture screen helped; reference 19 used seed 73. Garment screens included source-texture initialization; some invented trousers/limbs. All four reference-18 contact repairs left seams, so it was quarantined. V5 was superseded.
V6 SDXL / 42 trials	Local web/garment, whole-scene contact, detailed local garment, fabric-only and warped-fabric template repairs were screened. Whole-scene attempts spoiled contact shadows; local/fabric variants blurred anatomy or produced flat cloth. The 14 warped-fabric candidates were user-rejected. One local reference-2 result was only an intermediate selection.
V6 joint editor / 12 calls	Joint torso and both-sleeve edits from the built-in image editor were selected upstream. Seeds/model internals were not exposed. Their results survive as frozen source pixels, not as a reproducible seeded operation in this repository.
V7 final source correction	Three reference-2 geometry trials preceded a selected local rim/root editor repair, retaining 6,088 original shaft pixels and restoring color after registration. All 20 V7 source pairs were user-accepted, including reference 18. Four proposed SDXL repairs were not run because the GPU was busy. Detector-angle agreement was a diagnostic, not anatomical ground truth.
Final production choice	Freeze the 20 accepted pairs. Change only a guarded shared desk, then apply deterministic shared white balance and exact context masks. This avoids rerunning hand edits, segmentation and clothing repair for each appearance variant. The original accepted cohort is still separate from a new run of the clean code.

Project steps / where we are and what is next

The immediate deliverable is a clean generation baseline. Scientific evaluation is a separate phase with its own frozen cohort and protocol.

Completed / research and stimuli	Earlier model/control/repair comparisons established the image-building approach. V7 source pairs were approved. The original expansion produced 480 families / 2,880 PNGs and received user acceptance with the reference-17 clothing flag. The historical manifest preserves each accepted image hash.
Completed / clean repository	The 20 local mask/weight/preview sets now use correct crop coordinates; existing hash records and the recipe identity were updated. All 220 local source/input files pass validation. No source code was changed for this correction. Historical images, original production inputs and acceptance hashes are preserved. Full data and the large review PDF remain local.
Next 1 / validate clean CUDA run	Use the corrected local masks and a fresh output folder. Set up the pinned Linux CUDA environment and three models; run preflight and ten canaries. Inspect texture, the smaller contact boundary and all six images per family. Record approval before any full corrected run. No corrected-mask GPU sampling has been run.
Next 2 / freeze evaluation cohort	Select the historical accepted cohort or a newly accepted run explicitly. Audit common upstream source lineage before splitting; keep all six conditions and 24 appearance variants together, merging related references. Repeat analyses excluding reference 17: 456 families / 2,736 PNGs. Define static A-to-A and movement A-to-B timing before extracting features.
Next 3 / representation baselines	Compare CLIP and DINO representations with hand-only and simple low-level image baselines. Use grouped held-out sources and a preregistered body-context x movement contrast; report uncertainty by source group. Inspect desk, lighting and clothing confounds before interpreting model effects as motor-related.
Next 4 / motor-response model	Confirm the intended human motor-evoked-potential target and its stimulus protocol: the older report mentions thumb motion, while the current stimuli use index movement. Then establish a minimal spiking-network baseline, compare interactions on held-out source groups and document mismatches. Current image acceptance does not establish physiological replication.

Evidence scope: archived trial ledgers and saved outputs, the TRDP-I report, frozen V7 manifest, production recipe and the 11 October acceptance receipt. Earlier experiments are local history; their unused code and model caches are intentionally absent from the public repository.